Predicting VA Loan Approval Outcomes

A Machine Learning Classification and Fairness Analysis Using HMDA Data

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DSCI 7000 Capstone | Hawaii Pacific University

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339,211 applications analyzed

THE PROBLEM

A benefit designed to open doors

The VA home loan program was built as an earned benefit: zero down payment, no private mortgage insurance, more forgiving underwriting. Yet applications are still denied, and decades of research on conventional lending documents approval disparities across demographic groups.

If the program works as intended, **approval should turn on financial factors alone.**

Both possibilities are worth testing rather than assuming. That is the tension this study examines.

8.16%

of VA purchase applications denied

2nd layer

underwriting is increasingly automated

Models trained on past decisions learn whatever is in them, including unfair patterns.

RESEARCH QUESTIONS

Four questions, one dataset

RQ1 Which characteristics most strongly predict VA loan approval?

RQ2 Do nonlinear models beat a linear baseline, and what patterns explain the difference?

RQ3 Do models show disparate outcomes across groups, and how much is explained by underwriting?

RQ4 Does the Hawaii market differ from national patterns?

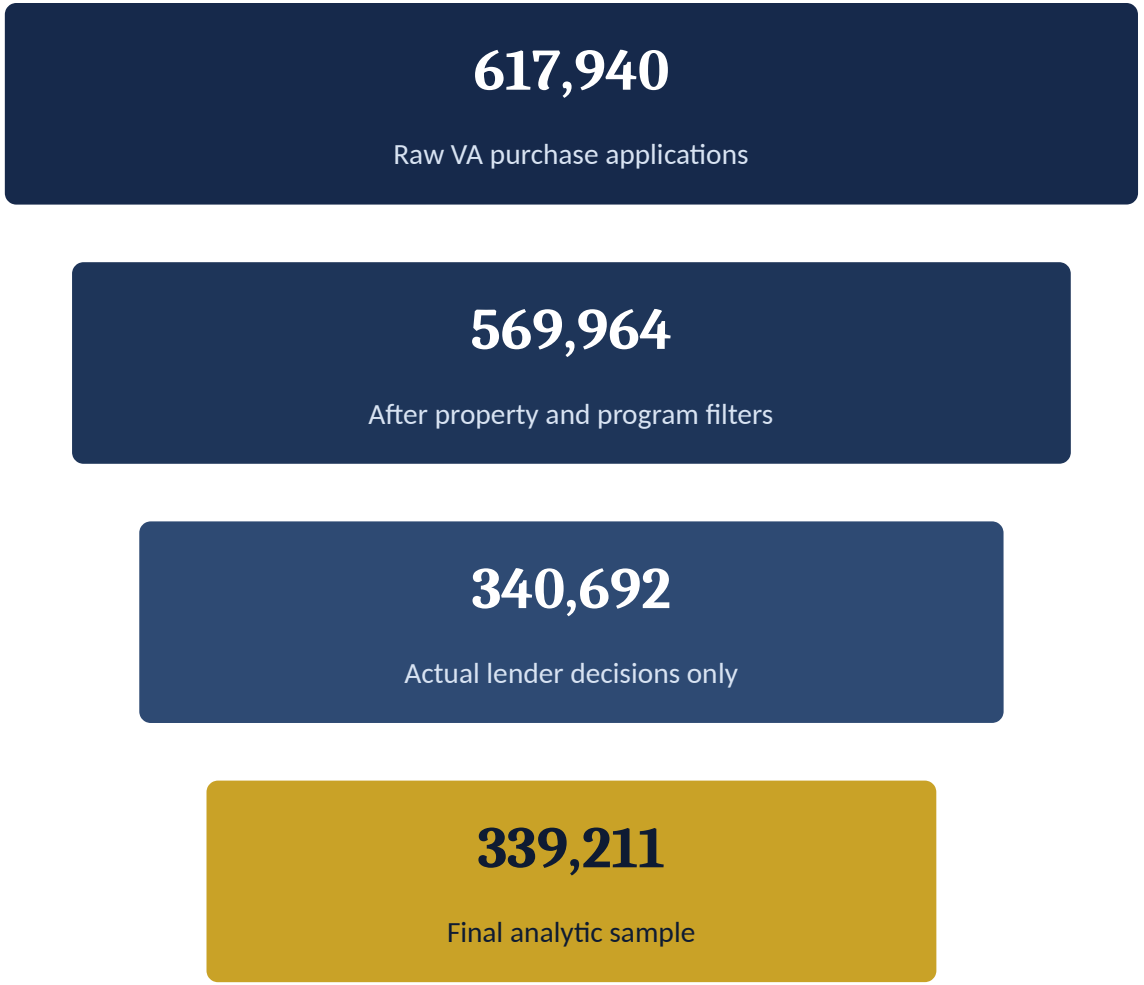
DATA

HMDA: federally mandated, publicly reported

Home Mortgage Disclosure Act loan-level records, 2025 filing year, downloaded from ffiec.cfpb.gov. Collected under federal statute and published by the CFPB and FFIEC. Public and de-identified, so no IRB approval required.

Overall approval rate: 91.84%

The largest single exclusion removed 229,272 records that were withdrawn, incomplete, or purchased loans. None represent a lender decision on creditworthiness.



METHODS

Fourteen fields had to be removed first

HMDA contains many fields that only exist once a loan is made. A model given these appears highly accurate while learning nothing about the approval decision.

1

Structurally missing for denials

Interest rate, rate spread, total loan costs, points and fees, origination charges, discount points, lender credits

A model can spot a denial just by detecting the blank.

2

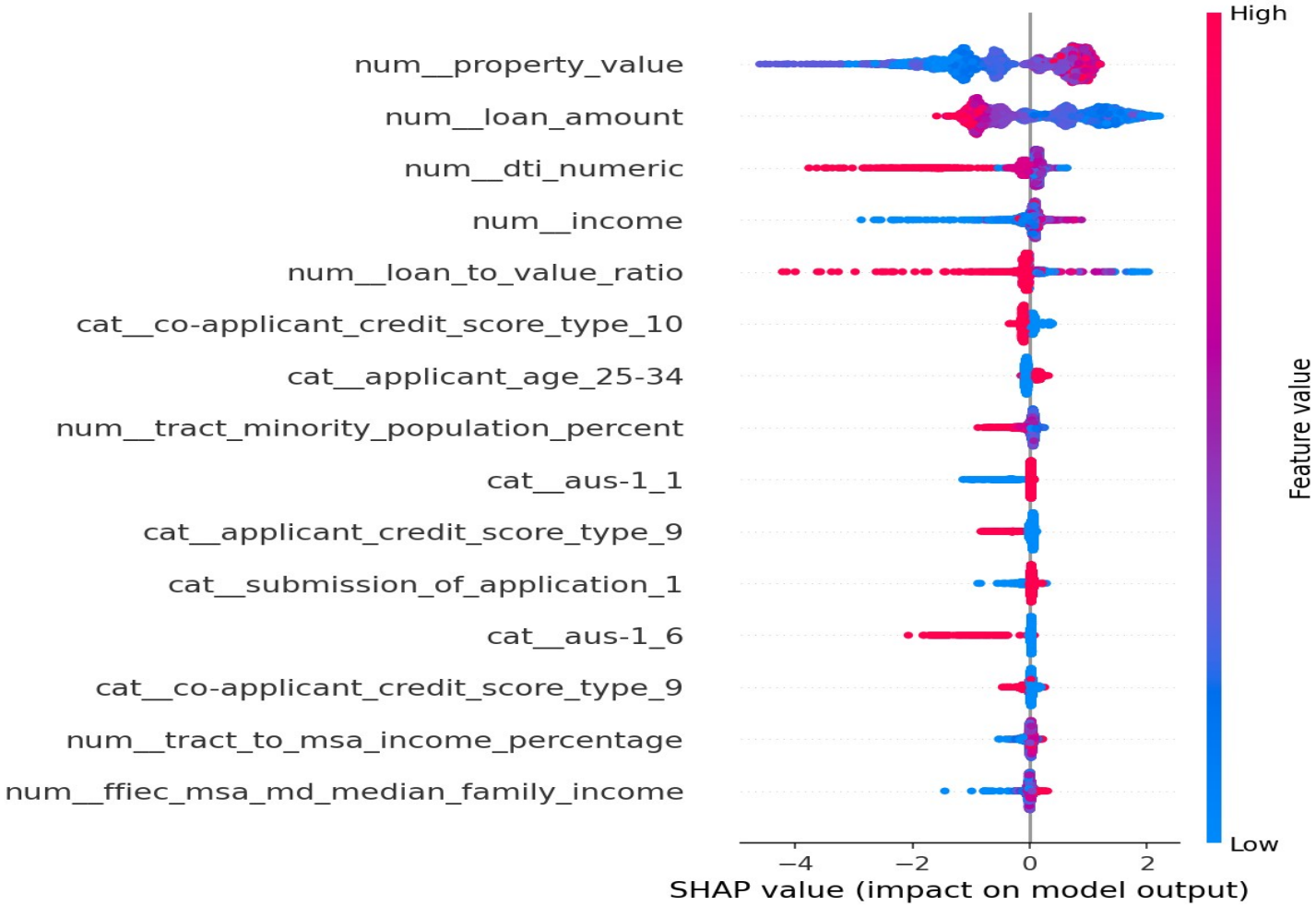
Outcome encoded in the category

HOEPA status, initially payable to institution, denial reason, purchaser type

HOEPA codes 1 and 2 show 100% approval; code 3 shows 23.6%. Univariate AUC 0.987.

RQ1

Property value and loan size dominate



Transaction size outweighs ratio-based risk

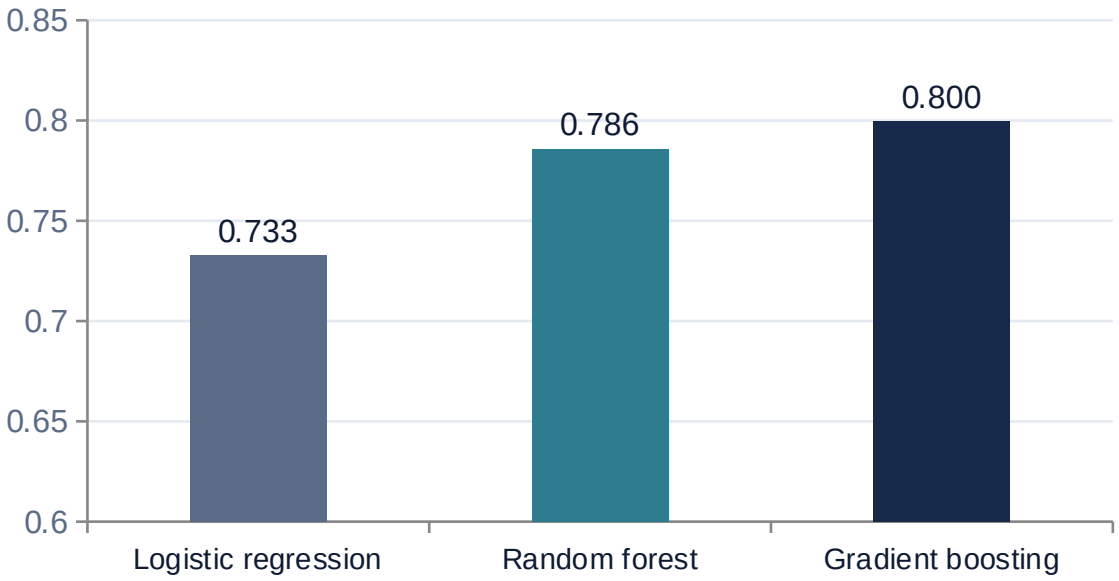
Property value (0.976) and loan amount (0.892) carry far more weight than DTI (0.220).

Neighborhood context contributes little

Tract minority population registers 0.077, an order of magnitude below the financial variables.

RQ2

Nonlinear models capture real structure



AUC-ROC by model

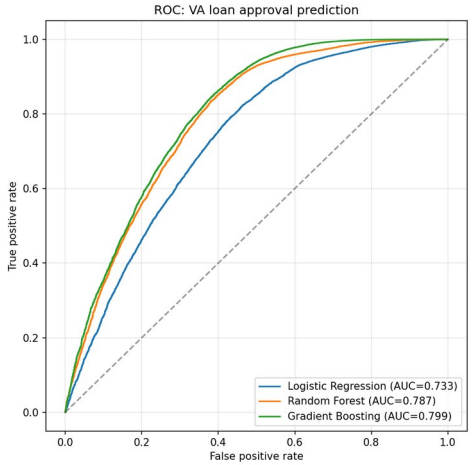
+0.067

AUC gain over the linear baseline

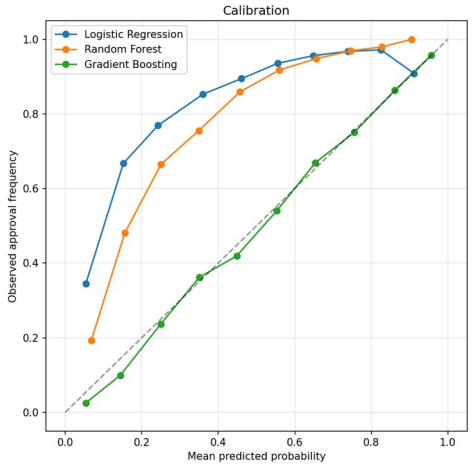
−72%

reduction in Brier score, 0.192 to 0.054

ROC curves

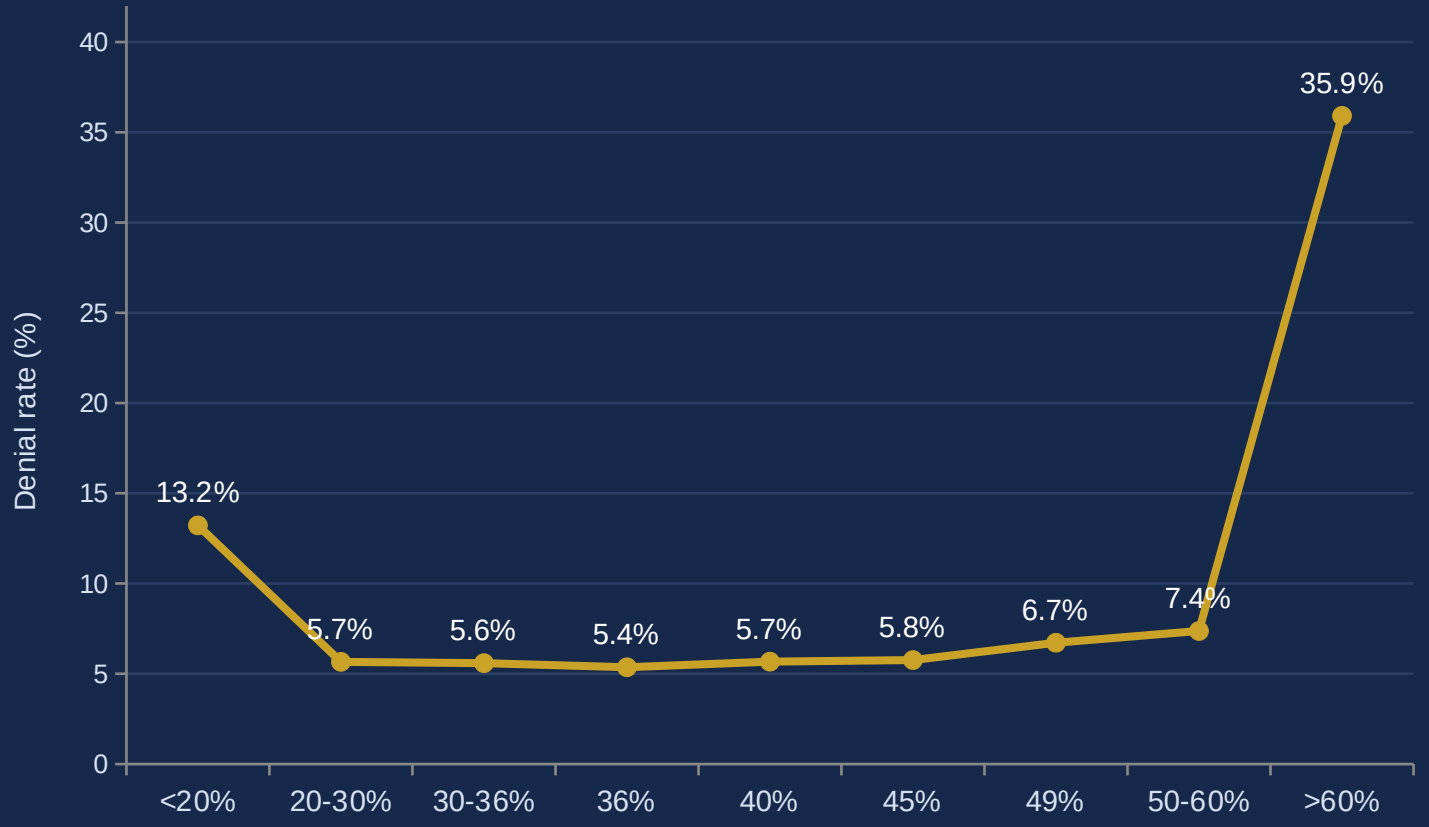


Calibration



RQ2 | PRINCIPAL FINDING

VA lending does not have the 50% DTI cliff



+0.7 pp

at the 50% threshold

+28.5 pp

above 60% DTI

Published research reports a 15 to 17 point jump at 50% for the general mortgage market. VA does not follow it.

Why VA underwriting behaves differently

Residual income, not a DTI ceiling

VA guidelines flag ratios above 41% for extra scrutiny but permit higher when residual income is adequate.

The constraint sits where conventional lending has already stopped

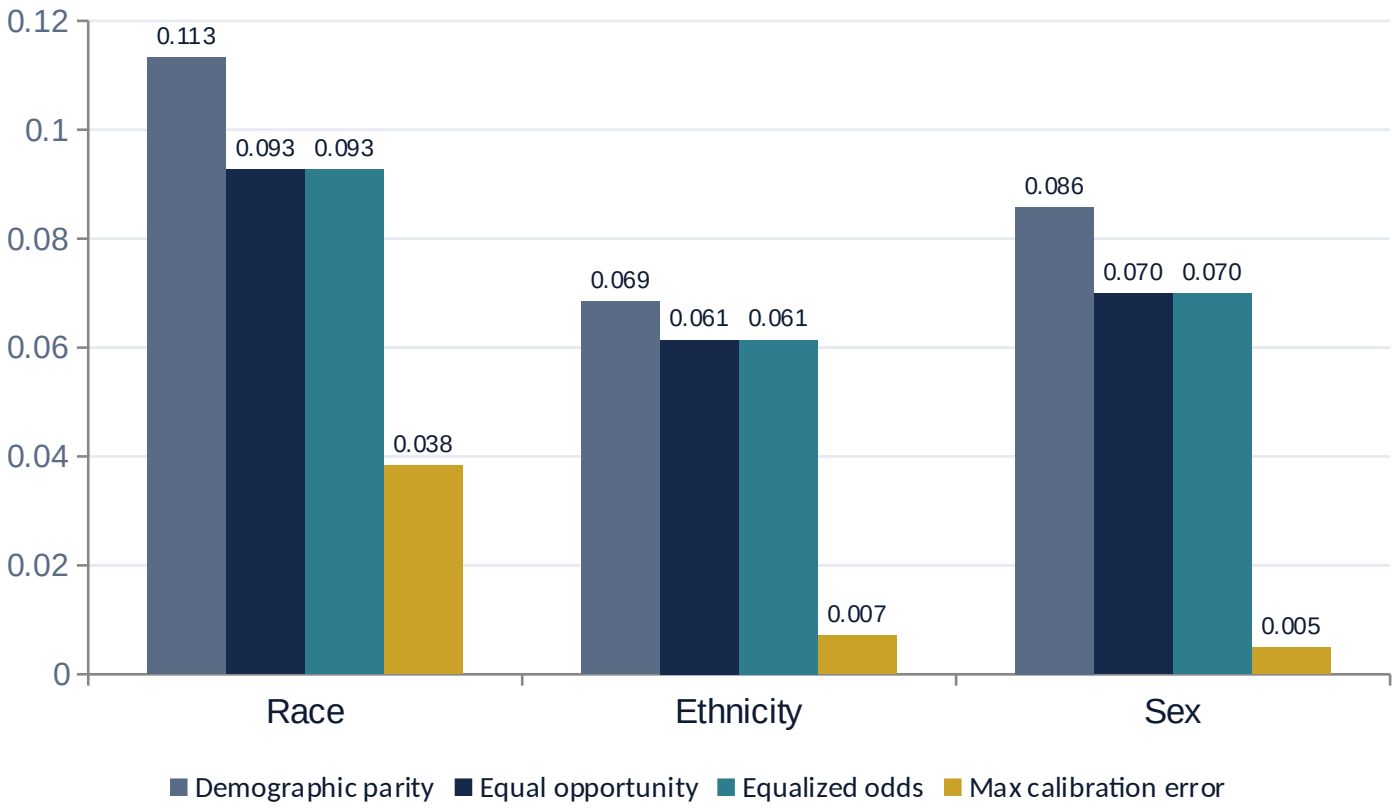
The binding threshold appears near 60%, well past the conventional market's cliff.

Low DTI is not simply protective

Applicants below 20% are denied at 13.22% versus 5.66% for the 20-30% band. A linear model would invert this.

Practical consequence: guidance calibrated to conventional lending understates a VA applicant's odds between 50% and 60% DTI.

Four fairness metrics, measured separately



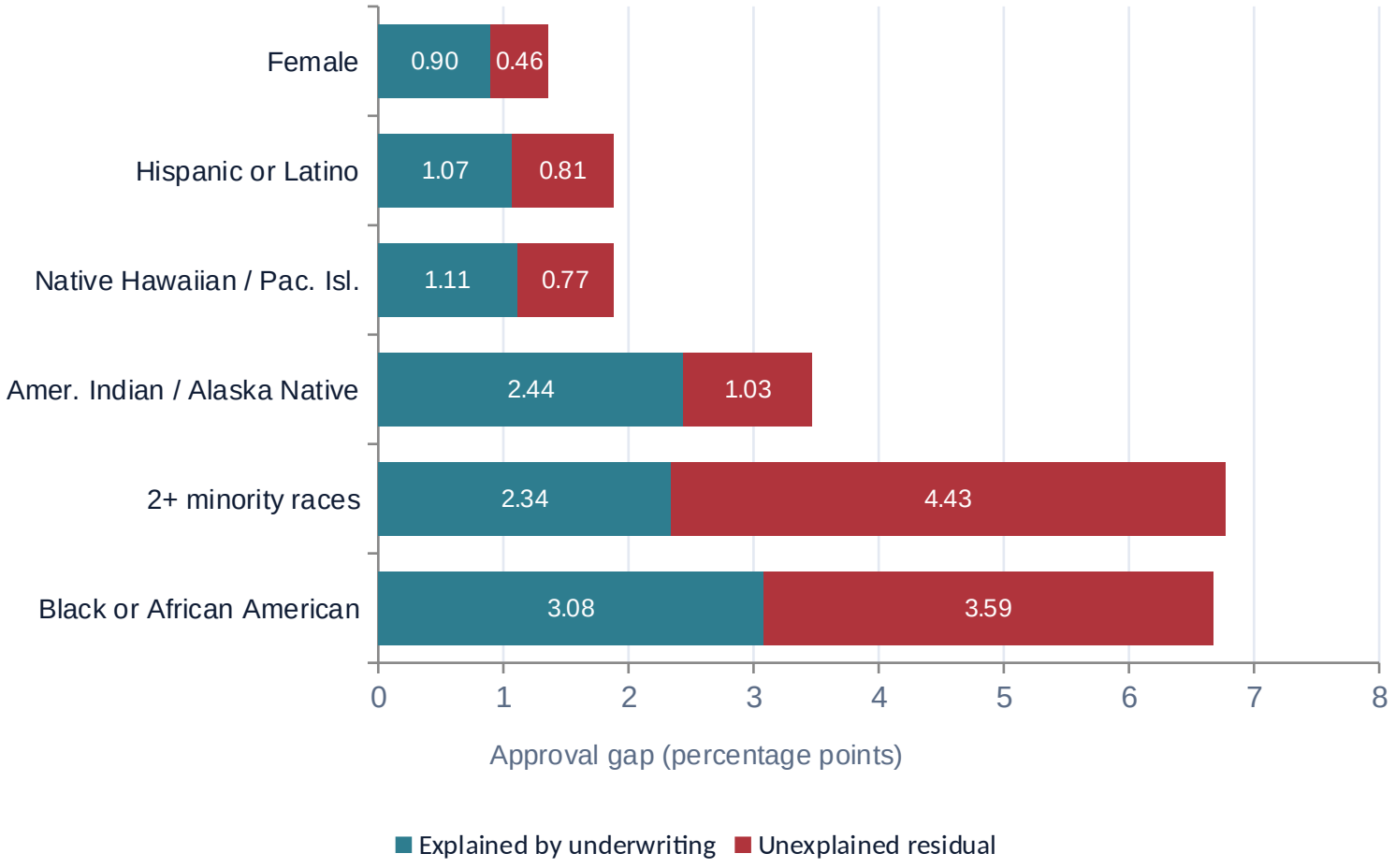
Parity gaps exceed opportunity gaps

The expected signature when groups differ in their underlying underwriting characteristics.

No single fairness number

The metrics conflict by design. Choosing one would amount to choosing a conclusion.

How much of each gap does underwriting explain?



46.1%

of the Black-White approval gap is explained by underwriting variables, leaving 3.59 points unexplained.

94.1%

of the Asian-White gap is explained. The gap is both small and almost entirely attributable to financial factors.

What these residuals do not establish

The public HMDA file does not contain credit score.

It is the single most influential variable in mortgage underwriting. Its absence means the controls in this analysis are incomplete by construction, and it is why the models plateau near AUC 0.80.

1

Residuals are upper bounds

Research using confidential data with credit scores found 12-point raw gaps shrink to 1-2 points once fully controlled.

2

Explained does not mean neutral

Differences in DTI and LTV have their own histories. The decomposition answers a narrower question than the word suggests.

3

Statistical disparity is not unfairness

It is a finding that justifies examination with richer data, not a verdict.

RQ4

Hawaii: the pattern replicates, the sample is thin

2,217

applications after screening

92.74%

approval rate, slightly above the national 91.84%

+18.1 pp

denial jump above 60% DTI, same direction as national

161

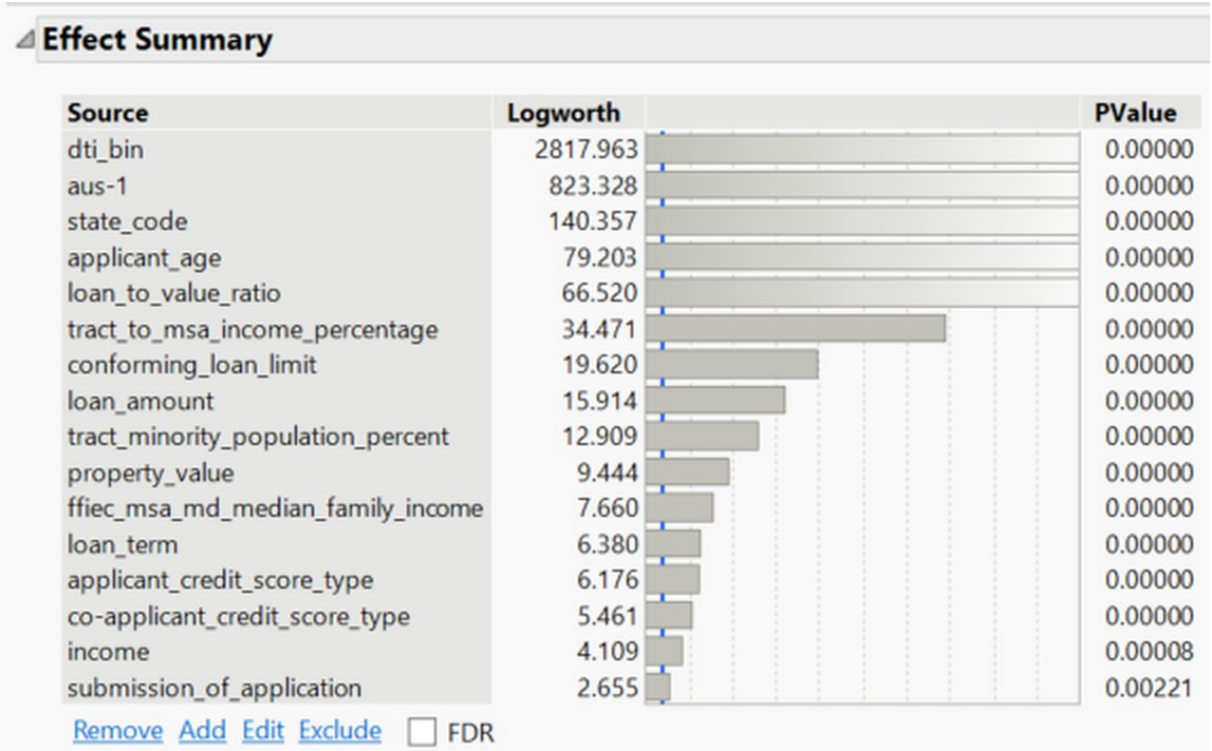
denials total, too few for group fairness metrics

Reported as descriptive corroboration, not an independent fairness audit. With 161 denials spread across nineteen DTI bins and several demographic groups below the minimum sample threshold, group-level fairness metrics could not be estimated reliably.

VERIFICATION

Two tools, same conclusion

The logistic baseline and the DTI analysis were rebuilt independently in JMP Pro from an identical exported dataset, so any discrepancy would reflect method rather than data.



Effect Summary

Source	Logworth	PValue
dti_bin	2817.963	0.00000
aus-1	823.328	0.00000
state_code	140.357	0.00000
applicant_age	79.203	0.00000
loan_to_value_ratio	66.520	0.00000
tract_to_msa_income_percentage	34.471	0.00000
conforming_loan_limit	19.620	0.00000
loan_amount	15.914	0.00000
tract_minority_population_percent	12.909	0.00000
property_value	9.444	0.00000
ffiec_msa_md_median_family_income	7.660	0.00000
loan_term	6.380	0.00000
applicant_credit_score_type	6.176	0.00000
co-applicant_credit_score_type	5.461	0.00000
income	4.109	0.00008
submission_of_application	2.655	0.00221

Remove Add Edit Exclude FDR

JMP Effect Summary, 16 effects, n = 268,569

2,817.96

logworth for DTI in JMP, more than 3x the next effect. Python found the same variable dominant through SHAP.

Zero sign reversals

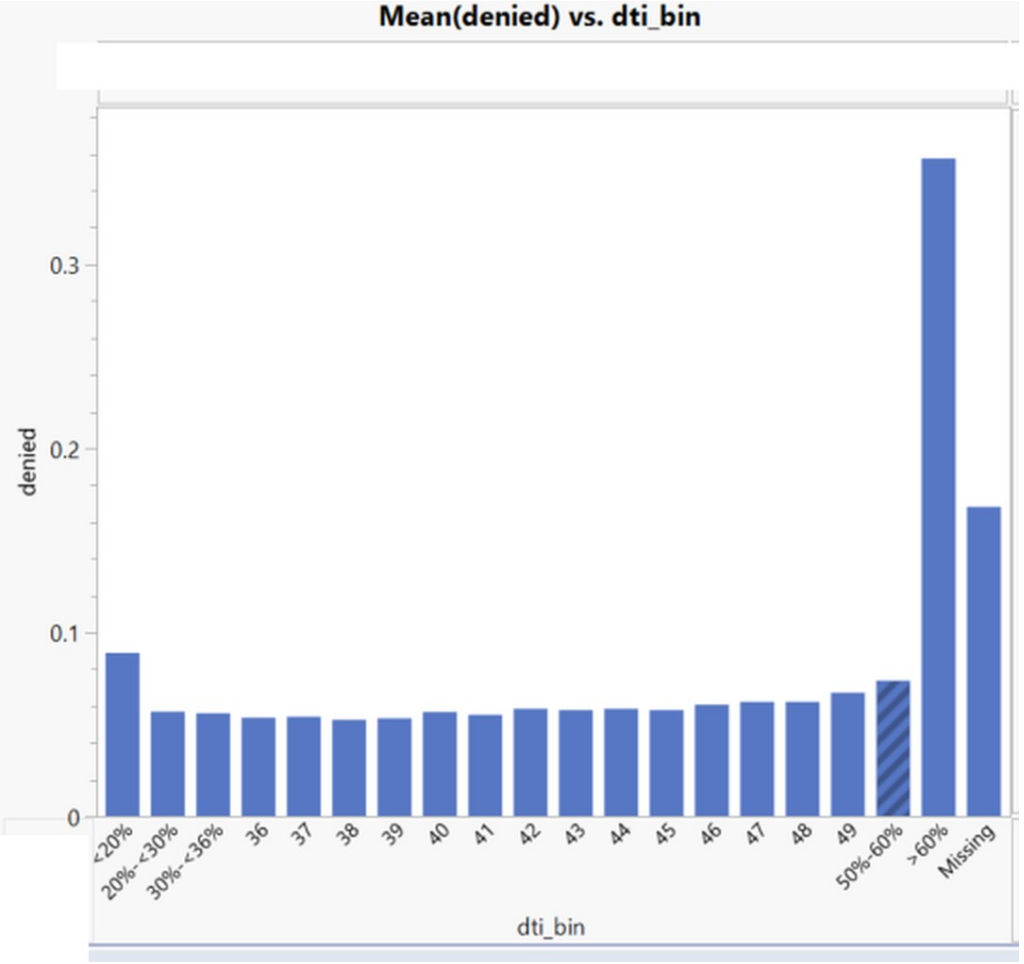
Geographic estimates agree within ~4%.

Different rankings, same answer

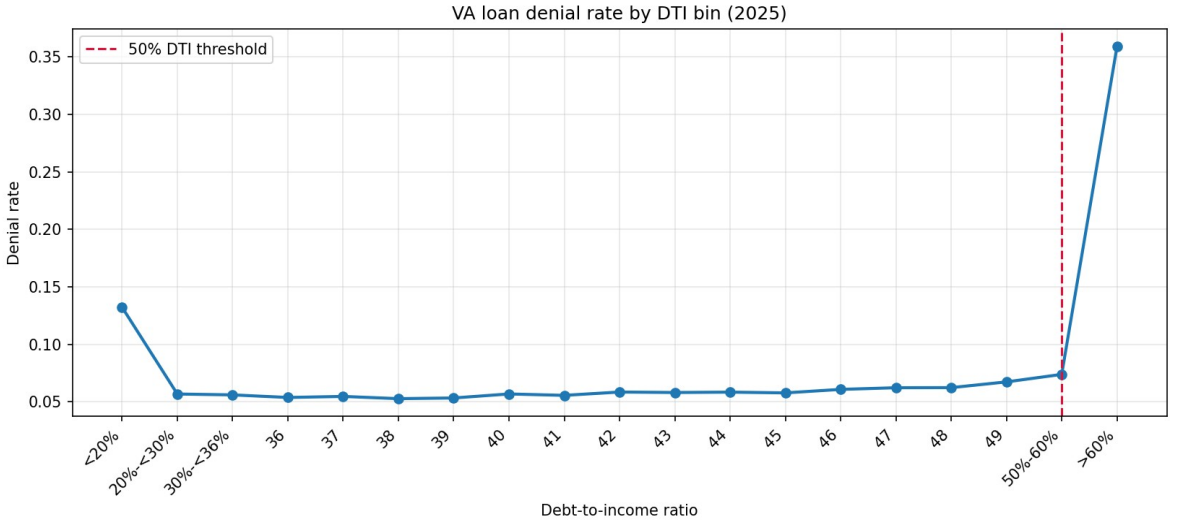
JMP ranks by statistical significance, Python by feature attribution. Both put DTI first.

VERIFICATION

The finding reproduces independently



JMP Graph Builder



Python / matplotlib

Same shape, same magnitudes: elevated below 20%, flat through the middle, cliff above 60%.

CONCLUSIONS

What this study establishes

1

Nonlinear models capture real structure

AUC 0.733 to 0.799; Brier score cut by more than 70%.

2

VA underwriting follows a different risk logic

The conventional 50% DTI cliff is nearly absent. The binding constraint sits above 60%, and the relationship reverses below 20%.

3

Underwriting explains 35% to 94% of demographic gaps

Residuals remain, but without credit score they are upper bounds, not evidence of discrimination.

Verified independently in Python and JMP Pro. Next step: replicate with credit score data to convert upper-bound residuals into interpretable estimates.

PRACTICAL IMPLICATIONS

Who should do what with this

Veterans and their agents

A VA applicant at 52% DTI faces a 7.37% denial rate, close to one at 45%. Conventional guidance understates their odds.

Lenders and regulators

Fairness audits of VA models need underwriting controls before any disparity is interpreted.

Model developers

Do not assume models calibrated on conventional lending transfer to VA underwriting.

Researchers

Replicate with credit score; test whether the 60% discontinuity is stable across years.

Mahalo

Data: HMDA 2025 national loan-level file, ffiec.cfpb.gov | Analysis: Python (scikit-learn, SHAP) with JMP Pro triangulation

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